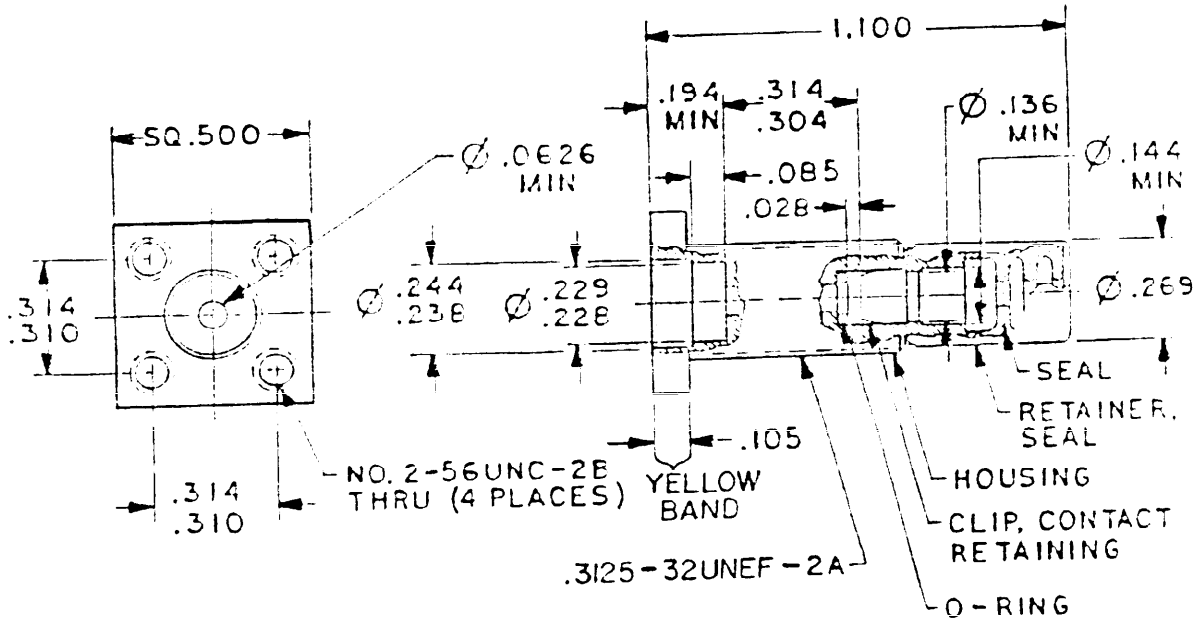
MILITARY SPECIFICATION SHEET

CONNECTOR, FIBER OPTIC, RECEPTACLE, FLANGE MOUNT ACTIVE (USED WITH MIL-T-29504/4 PIN TERMINUS, SIZE 16)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83522.



Inches	mm	Inches	mm
.028	0.71	.238	6.05
.085	2.16	.244	6.20
.105	2.67	.269	6.83
.136	3.45	.304	7.72
.144	3.66	.314	7.98
.194	4.93	.500	12.70
.228	5.79	1.100	27.94
.229	5.82		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. TO can sizes 47, 18, 46, 52.
4. Unless otherwise specified, all dimensions are maximum.

FIGURE 1. Dimensions and configuration.

FSC 6060

AMSC N/A

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REQUIREMENTS:

Metals: The plug shall be of corrosion resistant steel per QQ-S-763, class 303 or nickel plated brass per QQ-B-626. Other metal components shall be of corrosion resistant steel per QQ-S-763, class 303; nickel plated beryllium copper per QQ-B-530 or nickel plated brass per QQ-B-626.

Dimensions and configuration: See figure 1.

Optical requirements:

Number of optical termini: One.

Weight: 4 grams maximum.

Polarization: Not applicable.

Safety wire holes: Not required.

Force to engage and disengage:

Longitudinal force: 1 pound maximum.
Torque: Not applicable.

Terminus retention force: 24 pounds minimum.

Coupling proof torque: Not applicable.

Temperature: -65°C to +150°C.

Ozone exposure: Not applicable.

Part or Identifying Number: M83522/10-01.

Device accessories: Each device shall be supplied with one set of assembly instructions, and the assembly instructions shall include a listing of the epoxies specified, and a cleaning procedure statement.

Cleaning procedures: Only lint free lens tissue should be used for cleaning and drying of the terminus. It is particularly important when the terminus is dried that no residue of tissue lint remains on the optical surface. In addition, water should only be used as a secondary cleaner if isopropyl alcohol is unavailable. Many contaminants are insoluble in water and would remain after cleaning; therefore, use of water as the cleaning solvent should be avoided whenever isopropyl alcohol is available.

NOTEs: This receptacle mates with MIL-T-29504/4 terminus.
Insertion and removal tools: As specified in MIL-I-81969/14.

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TABLE III. Qualification inspection.

Inspection	Optical tests		Requirement paragraph	Test method paragraph
	Change in coupling loss 1/ 4.7.4	Signal discontinuities 4.7.5		
Group I (All sample units)				
Visual and mechanical inspection			3.4 thru 3.4.2 3.4.5 thru 3.4.1.1 3.6 and 3.7	4.7.1 and 4.7.1.1
Weight			3.4.3	4.7.2
Size			3.4.4	4.7.3
Group II (4 sample units)				
Temperature	A		3.5.3.2	4.7.11.2
Humidity	A		3.5.3.6	4.7.11.6
Submersion	A		3.5.3.7	4.7.11.7
Marking			3.6	4.7.1
Group III (4 sample units)				
Tensile loading 2/	B		3.5.2.1.1	4.7.6.1
Flex life	A		3.5.2.1.3	4.7.6.3
Twist	A		3.5.2.1.4	4.7.6.4
Maintenance aging	B		3.5.12	4.7.11.12 3/
Temperature cycling	B		3.5.3.3	4.7.11.3
Vibration		A	3.5.3.4	4.7.11.4
Shock		A	3.5.3.5	4.7.11.5
Ozone exposure			3.5.3.10	4.7.11.10
Marking			3.6	4.7.1
Group IV (4 sample units)				
Maintenance aging	B		3.5.3.12	4.7.11.12 3/
Pressure altitude	A		3.5.3.1	4.7.11.1
Temperature cycling	B		3.5.3.3	4.7.11.3
Humidity	B		3.5.3.6	4.7.11.6
Salt spray			3.5.3.8	4.7.11.8
Dust (fine sand)	B		3.5.3.9	4.7.11.9
Fluid immersion	B		3.5.3.11	4.7.11.11
Marking			3.6	4.7.1

1/ Allowable change is 1 dB max.

A - Indicates: Monitor change in optical throughput during test.

B - Indicates: Monitor change in optical throughput before and after test.

2/ Load to be 10 Kg.

3/ Test is accordance with method 2002, MIL-STD-1344, 25 cycles.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES
NASA - NA

User activities:
Navy - AS, MC, OS
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0074-2)

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